



Employee & Project Management Analytics

Dashboard Project Report

Tools

SQL + Power BI

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This report presents a comprehensive analysis of the **Employee & Project Management Analytics Dashboard**, developed using **Microsoft Power BI** and a **SQL Server** relational database. The system provides real-time visibility into workforce distribution, departmental budget utilization, project tracking, and individual employee performance across five business units — Engineering, Marketing, Sales, Human Resources, and IT.



Key Highlights:

- Total capital deployed across all departments stands at **\$1.288M**, with Engineering leading.
- **Sales department** generated the highest capital return of **\$1,16,000** against a project cost of \$1,50,000.
- **Human Resources** shows negative capital of **(\$25,000)** — budget overrun requires immediate attention.
- Project pipeline: **\$205K** completed + **\$365K** upcoming = **\$570K** total project investment.
- Dashboard supports drill-down filtering by Employee ID, Department, and Project Status.

■ **2. PROJECT OVERVIEW & OBJECTIVES**

#	Objective
1	Centralize workforce and project data into a single SQL database
2	Build an interactive Power BI dashboard with real-time filtering
3	Track project status (Completed vs Upcoming) across all departments
4	Analyze salary costs, project budgets, and capital performance per department
5	Enable management to make data-driven decisions on budget and staffing

3.1 Complete Employee Roster

Emp ID	First Name	Last Name	Job Title	Salary	Hire Date	Dept
1001	Liam	Brown	Software Engineer	\$85,000	2018-05-12	1
1002	Emma	Taylor	Marketing Manager	\$78,000	2020-03-10	2
1003	Michael	Johnson	Data Scientist	\$95,000	2016-09-01	3

4.2 Budget Allocation Visual

Department	Budget	Share	Visual
Engineering	\$1,200,000	34.8%	██
Marketing	\$800,000	23.2%	████████████████████████████
Sales	\$600,000	17.4%	████████████████
IT	\$450,000	13.0%	████████
Human Resources	\$400,000	11.6%	██████

Combined Total Budget: \$3,450,000 | Engineering leads at 34.8% share.

4.3 Power BI Dashboard — Financial Summary Table

Department	Goal	Project Cost	Salary Cost	Budget	2-Yr Budget	Capital
Engineering	Develop new products	\$1,10,000	\$1,60,000	\$24,00,000	12,00,000	\$7,70,000
Marketing	Increase brand awareness	\$1,15,000	\$1,58,000	\$16,00,000	8,00,000	\$3,69,000
Sales	Boost sales	\$1,50,000	\$1,67,000	\$12,00,000	6,00,000	\$1,16,000
Human Resources	Enhance employee engagement	\$1,05,000	\$1,60,000	\$8,00,000	4,00,000	(\$25,000)
IT	Improve IT infrastructure	\$90,000	\$1,51,000	\$9,00,000	4,50,000	\$58,000
TOTAL		\$5,70,000	\$7,96,000	\$69,00,000		\$12,88,000

■ **HR Alert: Human Resources capital is negative (\$25,000) — project + salary costs exceeded department budget. Immediate review required.**

5.1 Completed Projects (4)

Proj ID	Project Name	Budget	Start Date	End Date	Dept ID
204	CRM Integration	\$50,000	2022-01-20	2023-01-20	5
205	Brand Repositioning	\$70,000	2022-04-01	2023-04-01	2
208	Customer Support System	\$55,000	2022-12-15	2022-12-15	3
209	Market Research	\$30,000	2022-03-25	2023-03-25	4

5.2 Upcoming Projects (6)

Proj ID	Project Name	Budget	Start Date	End Date	Project Lead	Dept
201	Website Overhaul	\$60,000	2025-01-15	2025-06-30	Alice Green	5



upcoming_projects	PK: project_id	project_name, project_budget, project_start_date, project_end_date, dept_id, project_lead	Future projects
project_assignments	PK: assignment_id	employee_id (FK), project_id (FK)	M:M bridge table
Head_Shots	FK: employee_id	headshot_url	Employee photos

6.2 Master Analytics Query (CTE + Multi-JOIN)

```
-- CTE: Merge completed + upcoming projects with status label
WITH Project_Status AS (
SELECT project_id, project_name, project_budget,
'Upcoming' AS Status
FROM [upcoming_projects]
UNION ALL
SELECT project_id, project_name, project_budget,
'Completed' AS Status
FROM completed_projects
)
-- Main: Full join across all tables
SELECT
e.employee_id, e.first_name, e.last_name,
e.job_title, e.salary, d.Department_Name,
pa.project_id, ps.project_name,
ps.project_budget, ps.Status
FROM employees e
JOIN departments d ON e.department_id = d.Department_ID
JOIN project_assignments pa ON pa.employee_id = e.employee_id
LEFT JOIN Project_Status ps ON pa.project_id = ps.project_id;
```

Query Logic Explained: A CTE named *Project_Status* uses UNION ALL to combine both project tables into one with a Status label. The main SELECT then performs three JOINS — employees → departments → project_assignments — and a LEFT JOIN to project_status so that all employees are included even if their project has no matching record. This single output feeds every visual in the Power BI dashboard.



Figure 1 — Interactive Power BI Dashboard with Employee ID filter, Department slicer, and Status (Completed/Upcoming) filter. Shows \$1.288M total capital, project distribution donuts, department financial table, and project budget bar charts.

8. CONCLUSIONS & RECOMMENDATIONS

■ Budget Control	HR department has exceeded budget (–\$25,000 capital). Conduct an immediate budget audit and request supplementary funds or defer non-critical activities.
■ Top Performance	Engineering leads with \$7,70,000 capital on a \$12,00,000 budget (64% utilization). Their project management practices should be adopted as a company benchmark.
■ Project Readiness	6 upcoming projects worth \$365,000 are scheduled for 2025. Confirm project leads (Alice Green, Carol Black, Eva Brown etc.) have adequate team support.
■■■ Salary Review	Salary range spans \$68,000–\$95,000. A structured compensation framework will improve transparency, retention, and inter-department equity.
■ Highest Investment	Mobile App Development (\$70K) and Product Launch (\$80K) are the largest upcoming bets — assign senior project oversight and milestone tracking.
■ Dashboard Adoption	Distribute dashboard access to all department heads. Train them to use Employee ID and Status filters for real-time project performance monitoring.



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